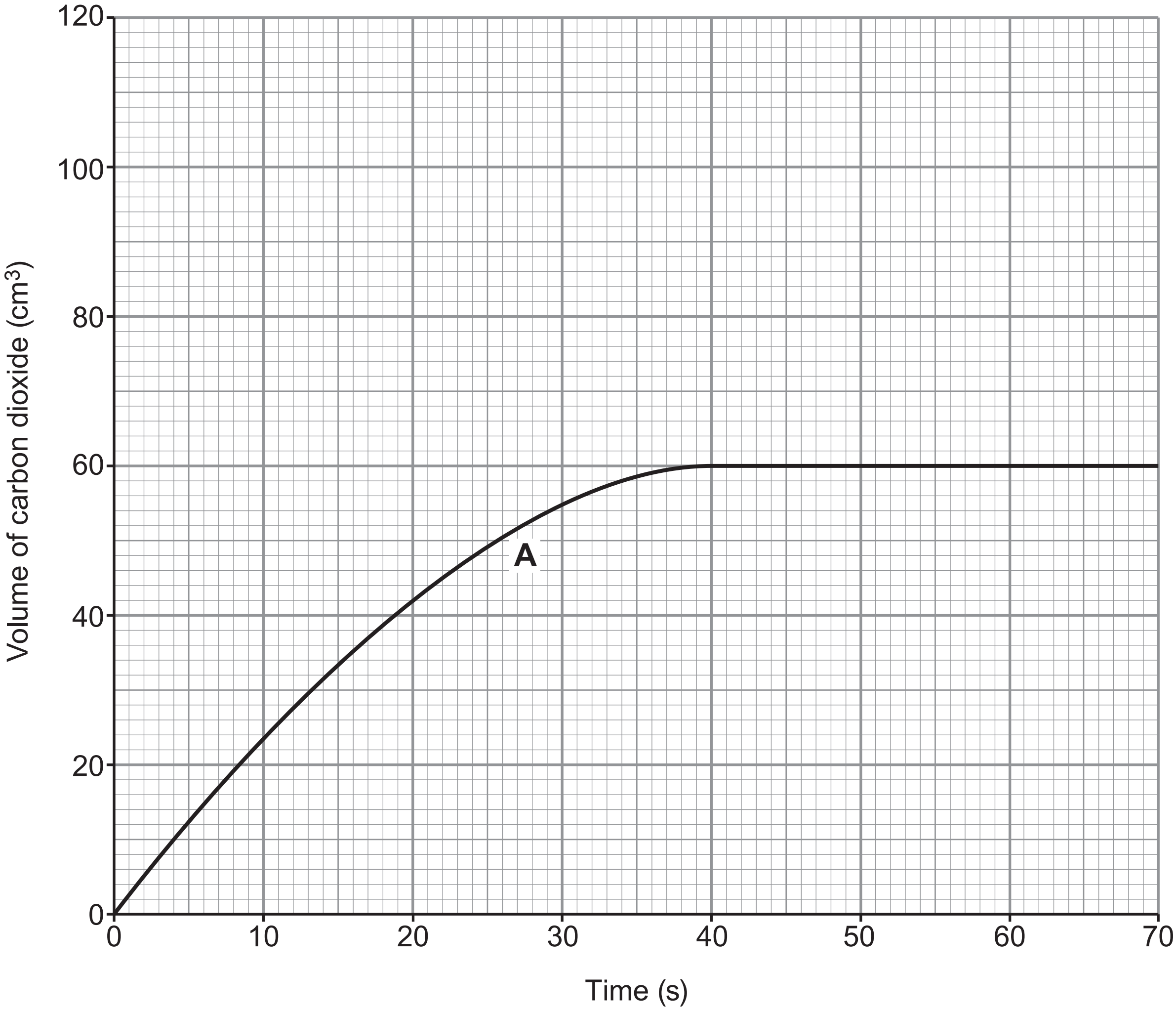
Examiner only

5. Marble chips (calcium carbonate) react with dilute hydrochloric acid to produce carbon dioxide gas.

Graph **A** shows the volume of carbon dioxide produced during the reaction between 0.25 g of marble chips and **excess** dilute hydrochloric acid at 30 °C.



(a) State why no more marble chips remain when the reaction stops. [1]

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(b) The original experiment was repeated at 60 °C.

On the grid draw the curve you would expect to see for the production of carbon dioxide at this temperature. [1]

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(c) The equation for the reaction between calcium carbonate and dilute hydrochloric acid is as follows.

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

Calculate the mass of calcium chloride produced when 7.8 g of calcium carbonate is reacted with excess acid. [2]

$M_r(CaCO_3) = 100$ $M_r(CaCl_2) = 111$

Mass = ... g

(d) When the reaction was repeated with a different mass of calcium carbonate, it was found that 14.3 g of calcium chloride formed. This represents a yield of 53.7%.

Calculate the maximum mass of calcium chloride that could have been formed. [2]

Maximum mass = ... g

(e) Catalysts can be added to some chemical reactions in order to increase their rate.

Explain how a catalyst increases the rate of a chemical reaction. [2]

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